INTERIM FEDERAL SPECIFICATION

ENAMEL, SEMI-GLOSS, FOR METAL AND WOOD FURNITURE

This Interim Federal Specification was developed by the Army Materials and Mechanics Research Center, Watertown, Massachusetts 02172, based upon currently available technical information. It is recommended that Federal Agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a synthetic semi-gloss enamel for metal and wood furniture that may be applied by brushing, spraying or dipping. It can be used in areas covered by AIR POLLUTION REGULATIONS.

1.2 Classification.

1.2.1 Classes. Enamel covered by this specification shall be of the following classes as specified (see 6.2):

Class A - Air-drying.
Class B - Baking.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

O-S-642 - Sodium Phosphate, Tribasic, Technical; Anhydrous, Dodecahydrate and Monohydrate
P-S-591 - Soap, Laundry, Bar
TT-B-846 - Butyl Alcohol, Normal (Butanol) (for Use in Organic Coatings)
TT-P-143 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing and Marking of
TT-P-636 - Primer Coating, Alkyd, Wood and Ferrous Metals
TT-T-291 - Thinner, Paint, Volatile Spirits (Petroleum-Spirits)
TT-T-548 - Toluene, Technical

FSC 8010

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Federal Standards:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing
Fed. Std. No. 595 - Colors

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge from the Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 476 - Specifications for Titanium Dioxide Pigments
ASTM D 2088 - Determination of Low Concentrations of Lead in Paint

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Color. The color of the enamel shall be as specified in the contract or order (see 6.2). When tested as in 4.2.3, it shall match the standard color chip in Fed. Std. No. 595 for 21105 Red, 22203 Orange, 23655 Yellow, 24097 Green, 25102 Blue and 27043 Gold (see 6.5).

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3.2 Composition.

3.2.1 Pigment. Hiding pigments shall be chemically pure, free from extenders and have good color permanence. The titanium dioxide shall be rutile chalk resisting type conforming to types III or IV of ASTM D 476. Extender pigments shall be predominately siliceous matter. The enamel shall contain no mercury compounds and the lead content shall be a maximum of 0.5 percent of the non-volatile when tested as in ASTM D 2088.

3.2.2 Vehicle.

3.2.2.1 Class A, air-drying. The vehicle for class A air-drying enamel shall be a drying oil phthalic alkyd resin together with the necessary additions of suitable thinners and driers. Small amounts of anti-skinning agents, wetting agents, suspension agents, and anti-drier absorption agents may be added at the discretion of the manufacturer. The vehicle shall show a negative rosin test.

3.2.2.2 Class B, baking. The basic vehicle for class B baking enamels shall be composed of a short oil length phthalic alkyd resin of the baking type modified with a suitable amount of a butylated melamine formaldehyde resin together with the addition of compatible solvents. Resins of the urea formaldehyde type blends of urea formaldehyde and melamine formaldehyde and triazine formaldehyde resins of the benzoguanamine type are permitted in lieu of melamine formaldehyde resins if formulated to yield enamels with baked film characteristics equivalent to those obtainable from the prescribed basic vehicle. Small amounts of stabilizers, wetting and suspension agents may be added at the discretion of the manufacturer.

3.2.2.3 Class A, air-drying and class B, baking. The volatile solvents used shall conform to the following requirements by volume when tested as in 4.2.4.

(a) Aromatic compounds with eight or more carbon atoms except ethyl benzene: 8 percent maximum.
(b) Ethyl benzene and toluene: 20 percent maximum.
(c) Solvents with an olefinic or cyclo-olefinic type of unsaturation: negative test (see 6.4).
(d) Ketones: negative.
(e) Total of a + b: 20 percent maximum.

3.3 Quantitative requirements. The enamel shall comply with table I when tested as in 4.2.

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TABLE I. Quantitative requirements

Characteristics	Requirements	
	Minimum	Maximum
Total solids, percent by weight of enamel		
All colors except gold	50	--
Gold	42	--
Mercury compounds	--	Negative
Total lead, percent by weight of nonvolatile	--	0.5
Hiding power (contrast ratio)		
Red, orange, yellow	0.90	--
Green, blue, gold	0.98	--
Phthalic anhydride, percent by weight of nonvolatile vehicle:		
Class A	20	--
Class B	26	--
Drying oil acids, percent by weight of nonvolatile vehicle:		
Class A	35	--
Class B	22	--
Rosin, on isolated vehicle	Negative	
Water, percent by weight of enamel	--	0.5
Coarse particles and skins (retained on No. 325 standard sieve), percent by weight of pigment	--	0.5
Flash point, Pensky-Martens, °F.	75	--
60° specular gloss	30	50
Viscosity (package):		
Class A		
Krebs-Stormer, shearing rate 200 r.p.m.:		
Grams	125	175
Equivalent K.U.	67	77
Class B		
Krebs-Stormer, shearing rate, 200 r.p.m.:		
Grams	100	150
Equivalent K.U.	61	72
Fineness of grind	6	--
Drying time:		
Class A, air-drying, hours		
Set to touch	1/4	2
Dry hard	--	8
Full hardness	--	72
Class B, baking		
Dry hard at 300°F., or equivalent heat treatment, minutes	--	30
Full hardness, air-drying, after baking, hours	--	24
Film hardness (class B)		
Pencil hardness	2B	--

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3.4 Qualitative requirements.

3.4.1 Condition in container. A freshly opened full container of enamel, tested as in 4.2.9 shall be free from grit, seeds, skins, lumps, abnormal thickening, or livering and shall show no more pigment settling or caking than can be readily reincorporated to a smooth homogeneous state.

3.4.2 Storage stability. The enamel shall show no skinning when tested as in 4.2.10. After aging as specified in 4.2.10, the enamel shall show no livering, curdling, hard caking, or gummy sediment. It shall mix readily to a smooth homogeneous state, shall have a maximum viscosity of 115 K.U. and any skin formed shall be continuous and easily removed.

3.4.3 Dilution stability. When thinned as in 4.2.11, the enamel shall remain stable and uniform showing no precipitation, curdling, or separation. Slight pigment settling shall be permitted.

3.4.4 Suspension properties. The enamel shall redisperse to a smooth, homogeneous state when tested as in 4.2.12.

3.4.5 Brushing properties (class A only). The enamel, tested as in 4.2.13, shall brush satisfactorily in all respects and shall dry to a smooth, uniform film, free from seeds, runs, sags, or streaks.

3.4.6 Spraying properties. The enamel, tested as in 4.2.14, shall spray satisfactorily in all respects, and shall show no running, sagging, or streaking. The air-dried (class A) or baked (class B) film shall show no dusting, mottling, or color separation and shall present a smooth uniform finish free from seediness.

3.4.7 Odor. When tested as in 4.2.15, the odor of the wet paint and of the film at any interval of drying shall not be obnoxious or objectionable.

3.4.8 Flexibility. A film of the enamel, tested as in 4.2.16, shall withstand bending without cracking or flaking.

3.4.9 Knife test. A film of enamel, tested as in 4.2.17, shall be hard and tough and shall adhere tightly to the metal and primed surface. It shall be difficult to furrow off with the knife and shall not flake, chip, or powder. The knife cut shall show beveled edges.

3.4.10 Recoating. A film of enamel, tested as in 4.2.18, shall not be self-lifting and there shall be no evidence of film irregularity.

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3.4.11 Water resistance. A film of the enamel, tested as in 4.2.19, shall show no wrinkling or blistering immediately after removal of the panel from water. The enamel shall be no more than slightly affected when examined 2 hours after removal; and after 24 hours air-drying, the portion of the panel which was immersed shall be almost indistinguishable with regard to hardness and color from the portion which was not immersed and shall have a gloss not lower than 20.

3.4.12 Hydrocarbon fluid resistance. A film of the enamel, tested as in 4.2.20, shall show no wrinkling or blistering immediately upon removal of the panel. After 24 hours air-drying, the portion of the panel which was immersed shall be almost indistinguishable with regard to hardness and color from a panel prepared at the same time but not immersed and shall have a gloss not lower than 20. A tendency for the red No. 21105 to become milky in appearance shall be disregarded.

3.4.13 Resistance to cleaning compounds. A film of the enamel tested as in 4.2.21 shall show no wrinkling or blistering immediately after completion of the test. After 24 hours air-drying the portion of the panel tested shall be almost indistinguishable with regard to hardness and color from a panel prepared at the same time but not tested and shall have a gloss not lower than 20.

3.4.14 Accelerated weathering. A film of the enamel, tested as in 4.2.22, shall show no more than slight chalking (No. 8 method 6411 of Fed. Test Method Std. No. 141), a color change equivalent to a lightness index difference not exceeding 4 units, and a gloss not lower than 20.

3.4.15 Toxicity. The product shall contain no benzol (benzene), chlorinated solvents, or hydrolyzable chlorine derivatives.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.1.1 Sampling and inspection. Sampling and inspection shall be performed in accordance with section 1000 of Fed. Test Method Std. No. 141.

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4.2 Test methods.

4.2.1 *Test conditions. The routine and referee testing conditions shall* be in accordance with section 7 of Fed. Test Method Std. No. 141 except as otherwise specified herein.

4.2.2 The following tests shall be conducted in accordance with Fed. Test Method Std. No. 141 or as required in this specification. The right is reserved to make any additional tests deemed necessary to determine that the enamel meets the requirements of this specification. Film tests on class B enamel shall be made after the coating has air-dried 5 to 10 minutes and then baked for 30 minutes at 300°F.

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TABLE II. Index

Item	Test Method: Applicable method in Fed. Std. No. 141	Test Method: Paragraph of this specification giving further references	Paragraph of this specification giving requirements
Color	4250	4.2.3	3.1
Isolation of vehicle (supercentrifuge)	4032	--	--
Total solids	4041	--	Table I
Aromatic hydrocarbons	--	4.2.4.2	3.2.2.3
Olefinic and cyclo-olefinic compounds	--	4.2.4.3	3.2.2.3
Ketones	--	4.2.4.4	3.2.2.3
Phthalic anhydride	7014	--	Table I
Drying oil acids	7014 1/	--	Table I
Rosin	5031	--	Table I
Flash point	4293	--	Table I
Water	4082	--	Table I
Coarse particles and skins	4092	--	Table I
Hiding power (contrast ratio)	4122	4.2.5	Table I
Specular gloss	6101	4.2.6	Table I
Viscosity, Krebs-Stormer	4281	--	Table I
Fineness of grind	4411	--	Table I
Drying time	4061	4.2.7	Table I
Film hardness	--	4.2.8	Table I
Condition in container	3011	4.2.9	3.4.1
Storage stability	3021	4.2.10	3.4.2
Dilution stability	4203	4.2.11	3.4.3
Suspension properties	--	4.2.12	3.4.4
Brushing properties	4321, 2141	4.2.13	3.4.5
Spraying properties	4331, 2131	4.2.14	3.4.6
Odor	4401	4.2.15	3.4.7
Flexibility	6221	4.2.16	3.4.8
Knife test	6304	4.2.17	3.4.9
Recoating	--	4.2.18	3.4.10
Water resistance	6011	4.2.19	3.4.11
Hydrocarbon fluid resistance	6011	4.2.20	3.4.12
Resistance to cleaning compounds	--	4.2.21	3.4.13
Accelerated weathering	6152, 6122	4.2.22	3.4.14
Toxicity	--	4.2.23	3.4.15

1/ Class B drying oil acids must be extracted with petroleum ether.

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4.2.3 Color. In accordance with method 4250 of Fed. Test Method Std. No. 141, compare the specified color with the pigmented coating on the white carrara glass panel prepared for the hiding power test (4.2.5) and observe for compliance with 3.1.

4.2.4 Solvent analysis.

4.2.4.1 Separation of volatile portion. Pour about 15 grams of the enamel into a 50 ml. distilling flask. Add 10 ml. of tricresyl phosphate and several anti-bumping stones or Berl saddles. Fit a release valve into the mouth of the flask and attach a delivery tube to the side arm, extending into a receiver. The receiver consists of a test tube (20 x 50 mm.) with side arm for attaching to a vacuum pump. The glass delivery tube should reach 1-1/2 inches from the bottom of the tube. Immerse the receiver in a dry ice-acetone bath. Preheat a silicone oil bath to 180°C. Raise the oil bath until the oil reaches the sample level. Reduce the pressure slowly to 10 mm. of mercury. After all solvent has distilled, carefully release the vacuum using the valve that is connected to the distilling flask. Reserve the collected distillate for the aromatic solvent determination and the test for ketone, olefinic and cyclo-olefinic compounds.

4.2.4.2 Determination of aromatic hydrocarbons.

Apparatus: A gas chromatograph equipped with a thermal conductivity detector.

Column preparation: Two lengths of 1/4-inch copper tubing, 6-ft. and 18-ft. long, are packed with 35 percent N,N-Bis (2-cyanoethyl)formamide on 60- to 80-mesh Chromosorb P.

Operating conditions:	6-ft.	18-ft.
Detector cell temperature, °C.	300	300
Detector cell current, ma.	150	150
Injection port temperature, °C.	300	300
Helium flow at exit, cc./minute	175	110
Column temperature, °C.	125	100

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4.2.4.2.1 Total aromatic content - procedure A. Transfer precisely 3 ml. of distillate or thinner to a 25-ml. glass-stoppered volumetric flask and add exactly 0.3 ml. of high purity phenylcyclohexane. While cooling the graduate under tap water, add 15 ml. of 85 percent sulfuric acid slowly. After all the acid has been added, shake vigorously for 2 minutes and allow the layers to separate. Add sufficient 85 percent acid to force the top layer into the neck of the flask and then transfer most of the top layer to a micro-separatory funnel. Wash the distillate with 5 ml. portions of distilled water until all acid has been removed and reserve the washed solvent for chromatographic analysis. Install the 6-ft. column and follow the operating conditions described above. Inject about 5 microliters of the acid-treated sample and allow the chromatogram to develop until the internal standard, phenylcyclohexane, emerges.

$$\% \text{ total aromatic solvents, v/v} = \frac{A \times 10^{*} \times 1.07^{**}}{B}$$

where, A = area of aromatic solvent peaks
B = area of internal standard peak
* = percent of internal standard
** = detector response correction factor

NOTE: If the above determination exceeds 8 percent, continue with the following procedure:

4.2.4.2.2 Toluene and ethylbenzene - procedure B. Treat 3 ml. of solvent in the same manner as described in procedure A except substitute benzene for phenylcyclohexane. Install the 18-ft. column and follow the operating conditions described for that column. Inject about 3 microliters of sample and allow the chromatograph to develop until all of the xylene isomers appear. Purge the column by raising the column temperature to 120°C. After the high boiling materials emerge, reset the column temperature to 100°C. Calculate the percent of toluene and ethylbenzene as follows:

$$\% \text{ toluene, v/v} = \frac{(\text{area of toluene peak})(1.017)^{*}\ (10)^{**}}{(\text{area of benzene peak})}$$

$$\% \text{ ethylbenzene, v/v} = \frac{(\text{area of ethylbenzene peak})(1.054)^{*}\ (10)^{**}}{(\text{area of benzene peak})}$$

where, * is the correction factor for the detector response
** is the percentage of internal standard added

NOTE: Sensitivity of the instrument should be adjusted to keep peaks from running off the scale. Appropriate corrections must be made for changes in sensitivity when computing the peak areas.

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4.2.4.3 Test for olefinic or cyclo-olefinic compounds. Take 2 test tubes and place 2 drops of the distillate in each. Dissolve the first sample in 1 ml. of carbon tetrachloride and add 1 drop of 1 percent bromine in carbon tetrachloride. Shake and allow to set for 5 minutes. A positive test is indicated by the complete absence of yellow color when observed against a white background. Dissolve the second sample in 1 ml. of acetone and add 1 drop of 1 percent permanganate solution (1 gram of potassium permangante crystals in 95 ml. of acetone and 5 ml. of water). Shake and allow to set for 2 minutes. A positive test is indicated by the decolorization of the purple solution. The solvent is considered to fail the test for olefinic and cyclo-olefinic compounds if either of the above tests is positive (see 3.2.2.3 and 6.4).

4.2.4.4 Test for ketones.

4.2.4.4.1 Reagent. Two grams of 2,4-dinitrophenylhydrazine plus 4 ml. of concentrated sulfuric acid plus 30 ml. methanol (add slowly) plus 10 ml. water.

4.2.4.4.2 Procedure. Pipet 1 ml. of reagent into a 20 by 170 mm. test tube. Add 10 drops of distillate and shake for 30 seconds. A yellow precipitate or cloud in the reagent layer indicates the presence of ketones. Run a blank using 1 ml. of reagent and 10 drops of mineral spirits.

4.2.5 Hiding power (contrast ratio). Determine the contrast ratio in accordance with method 4122 of Fed. Test Method Std. No. 141. Use a film applicator that will deposit a 3-inch wide film with a dry film thickness of 0.0010 inch maximum. Air-dry class A enamels for 72 hours and bake class B enamels for 30 minutes at 300° ± 4°F. Determine the reflectance and verify the film thickness in the area in which the reflectance was measured. Calculate the contrast ratio and check for compliance with Table I.

4.2.6 Specular gloss. Drawdown the packaged enamel using a 0.002 inch (0.004 inch gap clearance) film applicator. Air-dry class A for 48 hours under referee conditions; bake class B at 300° ± 4°F. for 30 minutes and cool to room temperature. Test as in method 6101 of Fed. Test Method Std. No. 141 for compliance with Table I.

4.2.7 Drying time. Determine drying time in accordance with method 4061 of Fed. Test Method Std. No. 141 under referee conditions using a 0.002 inch (0.004 inch gap clearance) film applicator.

4.2.7.1 Full hardness. The film shall be considered to have reached full hardness when it is difficult to remove with a knife blade.

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4.2.8 Film hardness (class B only). Prepare a film as in 4.2.6. Strip the wood from the lead of a standard drawing pencil[1/] for a distance of 1/4 inch. With a rotary motion, square the point of the exposed lead against No. 400 carbide abrasive paper. Hold the pencil at approximately 45° and push forward against the film using a pressure just short of breaking the lead. Clean the mark with a soap or "art gum" eraser. Hold the plate at an oblique angle in strong light and examine for marring. Determine the softest pencil that will mar the film. Express the film hardness as the next softest pencil and check for compliance with Table I.

4.2.9 Condition in container. Determine package condition in accordance with method 3011 of Fed. Test Method Std. No. 141 and observe for compliance with 3.4.1.

4.2.10 Storage stability. Determine skinning after 48 hours in accordance with method 3021 of Fed. Test Method Std. No. 141 except use a 3/4 filled 1/2-pint, multiple friction top can. Reseal and age for seven days at 140°F. and observe for compliance with 3.4.2.

4.2.11 Dilution stability. For class A, reduce one part by volume of enamel with one part by volume of thinner conforming to TT-T-291, Type II grade A. For class B, reduce one part by volume of enamel with one part by volume of thinner conforming to Table III. Test according to method 4203 of Fed. Test Method Std. No. 141 for compliance with 3.4.3.

TABLE III. Thinner for Class B

Ingredient	Percent by weight
VM&P naphtha (8% max. aromatic)	65
n-Butyl alcohol (TT-B-846)	20
Toluene (TT-T-548)	15

4.2.12 Suspension properties. Reduce the enamel as in 4.2.11, except use 4 parts of enamel to 1 part of thinner and place 6 ounces of the reduced enamel in an 8-ounce glass jar. Allow the stoppered jar to remain undisturbed for 24 hours and then place the unopened jar on a paint shaker[2/] and agitate the contents for 20 seconds. Examine the contents for evidence of nonhomogeneity or undispersed pigment. Observe for compliance with 3.4.4.

4.2.13 Brushing properties (class A only). Apply the enamel as packaged, or reduced with not more than 5 parts of thinner conforming to TT-T-291, Type II grade A to 95 parts of enamel by volume using a 2-1/2 inch brush and observe for brushing properties in accordance with method 4321 of Fed. Test Method Std. No. 141 for compliance with 3.4.5.

1/Eagle drawing pencil or equal.
2/An apparatus of this type, powered by a 1/4 hp. motor operates at a rate of 1350 shakes per minute and is manufactured by Red Devil Tools, Irvington, N.J.

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4.2.14 Spraying properties. Reduce the enamel as in 4.2.12. Spray on a steel panel to a dry film thickness between 0.0009 to 0.0011 inch and observe for spraying properties in accordance with method 4331 of Fed. Test Method Std. No. 141 for compliance with 3.4.6. For referee test, use automatic application per method 2131 of Fed. Test Method Std. No. 141.

4.2.15 Odor. Test for odor in accordance with method 4401 of Fed. Test Method Std. No. 141 and observe for compliance with 3.4.7.

4.2.16 Flexibility. Determine flexibility in accordance with method 6221 of Fed. Test Method Std. No. 141. Apply a 2-inch wide film of enamel with a film applicator that will give a dry film thickness of 0.0009 to 0.0011 inch on a smooth finish steel panel prepared in accordance with method 2011 of Fed. Test Method Std. No. 141 using the aliphatic naphtha-ethylene glycol monoethyl ether mixture. The panel shall be prepared from new cold rolled carbon steel rust-free 0.010 ± 0.001 inch thick with a Rockwell 15-T maximum hardness of 82 and finished with a surface roughness of 8 to 12 microinches. Air-dry class A enamel in a horizontal position for 18 hours and then bake for 168 hours at 105° ± 2°C. (221° ± 4°F.). Bake class B enamel for 30 minutes at 300°F. and then bake for 168 hours at 105° ± 2°C. (221° ± 4°F.). Condition the panel for 1/2 hour under referee conditions. Bend over a 1/4 inch mandrel and examine the coating for cracks over the area of the bend for compliance with 3.4.8.

4.2.17 Knife test. Perform the knife test in accordance with method 6304 of Fed. Test Method Std. No. 141. Phosphoric acid etch a steel panel in accordance with method 2011, procedure B, of Fed. Test Method Std. No. 141. Mask one half the panel, and using a 0.002 inch (0.004 inch gap clearance) film applicator, draw down a 3-inch wide film of primer conforming to TT-P-636. Air-dry for 18 hours, remove masking, and using a 0.002 inch (0.004 inch gap clearance) film applicator draw down a 2-inch wide film of enamel over the entire length of the panel. Air-dry class A enamel for 72 hours and examine for compliance with 3.4.9. For class B enamels, bake the primer and topcoat each 30 minutes at 300°F. and condition for 24 hours at room temperature before testing.

4.2.18 Recoating. Prepare the enamel film as in method 4061 of Fed. Test Method Std. No. 141. Air-dry class A enamel for 24 hours under referee testing conditions. Bake class B enamel for 30 minutes at 300°F. and allow to return to room temperature. Apply a second coat crosswise to the first coat and air-dry or bake as before. Examine for compliance with 3.4.10.

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4.2.19 Water resistance. Draw down a film of enamel with a 0.002 inch (0.004 inch gap clearance) film applicator on two steel panels phosphoric acid etched as in method 2011, procedure B, of Fed. Test Method Std. No. 141. Air-dry class A enamel for 7 days and class B enamel for 24 hours after baking for 30 minutes at 300°F. Coat all exposed unpainted surfaces with wax or a suitable protective coating. Then immerse one panel in distilled water at 23° ± 1°C. for 18 hours in accordance with method 6011 of Fed. Test Method Std. No. 141. Immerse the second panel in boiling distilled water and allow the water to boil gently for a 15 minute immersion period. At the end of the test period, remove and examine for compliance with 3.4.11.

4.2.20 Hydrocarbon fluid resistance. Prepare a film of the enamel as in 4.2.19, except unpainted metal areas shall not be wax coated. Air-dry or bake the specimen for the specified time and then immerse in thinner conforming to TT-T-291, Type II grade A at 23° ± 1°C. for 4 hours in accordance with method 6011 of Fed. Test Method Std. No. 141. At the end of the test period, remove and examine for compliance with 3.4.12.

4.2.21 Resistance to cleaning compounds. Prepare four panels as in 4.2.19 except unpainted metal areas shall not be coated with wax.

4.2.21.1 Trisodium phosphate. Place one of the test panels in a beaker containing 2.5 inches of a 2 percent aqueous solution of trisodium phosphate (conforming to O-S-642) at a temperature of 50°C. (122°F.) for class A enamel and 82°C. (180°F.) for class B enamel. Allow the panel to remain immersed at the specified temperature for 1/2 hour. Remove the panel from the solution and wash thoroughly under running water using a soft sponge. Gently wipe dry with a soft cloth and examine for compliance with 3.4.13.

4.2.21.2 Soap. Place one of the test panels in a beaker containing about 2.5 inches of an aqueous solution of 2 percent yellow laundry soap (conforming to P-S-591) and 0.25 percent of sodium hydroxide. Leave immersed for 15 minutes. Remove the panel from the solution and heat the solution to 50°C. (122°F.) for class A enamel, and 82°C. (180°F.) for class B enamel. Place another panel, not previously tested, in the warm solution for 5 minutes, maintaining the temperature specified. Remove the panel from the solution and wash thoroughly under running water, using a soft sponge. Gently wipe dry with a soft cloth and examine for compliance with 3.4.13.

4.2.21.3 Alcohol. Place one of the test panels in a beaker containing about 2.5 inches of an aqueous solution of 50 percent ethyl alcohol. Allow to remain at room temperature for 15 minutes. Remove panel from the solution and wash thoroughly under running water with a soft sponge. Gently wipe dry with a soft cloth and examine for compliance with 3.4.13.

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4.2.22 Accelerated weathering resistance. Draw down a film of enamel on duplicate flat tin panels (method 2012 of Fed. Test Method Std. No. 141) with a 0.002 inch (0.004 inch gap clearance) film applicator. Air-dry class A enamel for 96 hours. Bake class B enamel for 30 minutes at 300°F. or equivalent heat treatment and air-dry for 24 hours. Measure the gloss and directional reflectance of the coating. Subject the coated panels to 168 hours of accelerated weathering in accordance with method 6152 of Fed. Test Method Std. No. 141 using a twin arc apparatus. Examine the exposed coating for chalking by rubbing with a piece of velvet or cheesecloth. Wash the exposed panel under running water with a soft sponge. Wipe off water with clean cheesecloth and let dry for 2 hours. Determine loss of gloss, and then measure the directional reflectance (method 6121 of Fed. Test Method Std. No. 141) on an unrubbed area of the exposed panel and determine the amount of color change, expressed as lightness index difference (ΔL), using method 6122 of Fed. Test Method Std. No. 141. Check for compliance with 3.4.14.

4.2.23 Toxicity. The manufacturer shall certify that the enamel contains no benzene (benzol), chlorinated solvents or hydrolyzable chlorine derivatives.

4.2.24 Packaging, packing and marking. The enamel shall be inspected for compliance with the packaging, packing and marking requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The enamel shall be packaged, packed and marked in accordance with TT-P-143. The level of packaging shall be A or C, and the level of packing shall be A, B, or C, as specified (see 6.2). The paint shall be furnished in 16 ounce pressurized dispensers or 1-quart or 1-gallon multiple friction top containers as specified (see 6.2).

6. NOTES

6.1 Intended use. The enamel covered by this specification is intended for use on metal and wood furniture. It will provide a finish of maximum washability and durability.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).
(c) Color required (see 3.1).
(d) Size of container required (see section 5).
(e) Selection of applicable levels of packaging and packing (see 5.1).

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6.3 The enamel covered by this specification should be purchased by volume, the unit being one U.S. liquid gallon of 231 cubic inches at 20°C. (68°F.).

6.4 The test for olefinic and cyclo-olefinic compounds will not be positive for solvents containing less than 1 percent of these compounds.

6.5 Color standards. At the present time Fed. Std. No. 595 does not contain semigloss color standards for 24097 Green and 27043 Gold but action is being taken to add these colors. For 24097 Green and 27043 Gold, the color standards shall be furnished by the procuring agency or obtained from General Services Administration, Federal Supply Service - FMSB, Washington, D.C. 20406.

MILITARY CUSTODIAN:

Army - MR

Preparing activity:

Army - MR

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

TT-E-001793(Army-MR), Enamel, Semi-Gloss, for Metal and Wood Furniture

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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